Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
2 P-only DeepONet10	**10k**	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	**long**	1.12	0.631	2.57e+06	2.60e+08	0.494	1.03e+09	7.18e+05	65.3	782	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.42	0.799	1.48e+06	3.29e+08	0.625	1.26e+09	9.10e+05	117	1.44e+03	0	Complete
4 T/P DeepONet10	**10k**	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	**long**	0.307	0.975	6.25e+08	1.28e+08	0.969	3.29e+10	5.42e+05	39.2	886	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.409	0.116	6.85e+08	3.71e+05	5.26e-03	1.56e+07	4.57e+03	146	2.09e+03	0	Complete
6 T/P/ϕ DeepONet10	**10k**	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	**long**	7.51e-06	1.68e-06	9.73e+03	0.602	4.05e-06	323	0.798	0.945	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0301	0.0167	102	650	0.0155	290	1.54e+04	176	3.15e+03	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.